1 Physics Period 1, Week 3

1. A rocket rises with constant acceleration to an altitude of 85 km, at which point its speed is 2.8 km/s.

   (a) What's its acceleration?

   (b) How long does the ascent take?

2. A car moving initially at 50 mi/h begins slowing at a constant rate 100 ft short of a spotlight. If the comes to a full stop just at the light, what is the magnitude of its acceleration?

3. You drop a rock into a deep well and 4.4 sec later hear a splash. How far down is the water? Neglect the travel time of sound.

4. You allow 40 min to drive 25 mi to the airport, but you're caught in heavy traffic and average only 20 mi/h for the first 15 min. What must your average speed be on the rest of the trip if you're to make your flight?

5. A jetliner touches down at 220 km/h and comes to a halt 29 s later. What's the shortest runway on which this aircraft can land?

6. After 35 min of running, at the 9 km point in a 10 km race, you find yourself 100 m behind the leader and moving at the same speed. What should your acceleration be if you're to catch up by the finish line? Assume that the leader maintains constant speed.

7. Two divers jump from a 3.00-m platform. One jumps upward at 1.80 m/s, and the second steps off the platform as the first passes it on the way down.

   (a) What are their speeds as they hit the water?

   (b) Which hits the water first and by how much?

8. A police radar's effective range is 1.0 km, and your radar detector's range is 1.9 km. You're going 110 km/h in a 70 km/h zone when your radar detector beeps. At what rate must you negatively accelerate to avoid a speeding ticket?